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Course/Section: CPE212-CPE31S1	**Date Submitted: 03/12/2026**
Instructor: Engr. Robin Valenzuela	**Semester and SY: 2nd Sem**

Midterm Skills Exam: Install, Configure, and Manage Log Monitoring tools

1. Objectives

Create and design a workflow that installs, configure and manage enterprise availability, performance and log monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

2. Instructions

1. Create a repository in your GitHub account and label it CPE_MIDEXAM_SURNAME.
2. Clone the repository and do the following:
   2.1. Create an Ansible playbook that does the following with an input of a config.yaml file and arranged Inventory file:
   2.2. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash) • Install Nagios in one host
   2.3. Install Lamp Stack in separate hosts (Httpd + Php,Mariadb)
3. Document all your tasks using this document. Provide proofs of all the ansible playbooks codes and successful installations.
4. Document the push and commit from the local repository to GitHub.
5. Finally, paste also the link of your GitHub repository in the documentation.

3. Output (screenshots and explanations)

2.1:

inventory.ini | ansible.cfg

ansible.cfg

```
[defaults]
inventory = inventory.ini
private_key_file = /home/isaac/.ssh/id_rsa
```

inventory.ini

```
[Isaac]
192.168.56.114

[Jhon]
192.168.56.118 ansible_user=jhonisaacalegre
```

inventory.ini | ansible.cfg | main.yml nagios/... | midterm.yml

midterm.yml

```yaml
---
- name: Install Elasticsearch on Ubuntu and CentOS hosts
  hosts: ubuntu:centos
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: elasticsearch
  roles:
    - Elastic

- name: Install Kibana on Ubuntu and CentOS hosts
  hosts: ubuntu:centos
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: kibana
  roles:
    - Elastic

- name: Install Logstash on Ubuntu and CentOS hosts
  hosts: ubuntu:centos
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: logstash
  roles:
    - Elastic

- name: Install Nagios on Ubuntu
  hosts: ubuntu
  become: true
  vars_files:
    - config.yaml
  roles:
    - nagios

- name: Install Apache and PHP on Ubuntu and CentOS hosts
  hosts: ubuntu:centos
  become: true
```

```yaml
    become: true
    vars_files:
      - config.yaml
    vars:
      lamp_component: web
    roles:
      - lamp

  - name: Install MariaDB on Ubuntu and CentOS hosts
    hosts: ubuntu:centos
    become: true
    vars_files:
      - config.yaml
    vars:
      lamp_component: db
    roles:
      - lamp
```

2.2:

Code:
Elastic Task:

```yaml
---
- name: Install Java for Elastic Stack on Ubuntu
 apt:
   name: "{{ elastic_java_package_ubuntu }}"
   state: present
   update_cache: yes
 when: ansible_facts['distribution'] == 'Ubuntu'


- name: Install Java for Elastic Stack on CentOS
 dnf:
   name: "{{ elastic_java_package_centos }}"
   state: present
```

```
    update_cache: yes
  when: ansible_distribution == 'CentOS'

- name: Add Elastic GPG key
  apt_key:
    url: "{{ elastic_repo_key }}"
    state: present
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Add Elastic repository
  apt_repository:
    repo: "{{ elastic_repo_url }}"
    state: present
    filename: elastic
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Add Elastic repository on CentOS
  yum_repository:
    name: elastic-8.x
    description: "{{ elastic_repo_description_centos }}"
    baseurl: "{{ elastic_repo_baseurl_centos }}"
    gpgcheck: yes
    gpgkey: "{{ elastic_repo_key }}"
    enabled: yes
  when: ansible_distribution == 'CentOS'
```

```yaml
- name: Install requested Elastic component
 apt:
   name: "{{ elastic_component }}"
   state: present
   update_cache: yes
 when: ansible_facts['distribution'] == 'Ubuntu'


- name: Install requested Elastic component on CentOS
 dnf:
   name: "{{ elastic_component }}"
   state: present
   update_cache: yes
 when: ansible_distribution == 'CentOS'


- name: Configure Elasticsearch network settings
 lineinfile:
   path: /etc/elasticsearch/elasticsearch.yml
   regexp: '^{{ item.key }}:'
   line: "{{ item.key }}: {{ item.value }}"
 loop:
   - { key: 'network.host', value: '0.0.0.0' }
   - { key: 'discovery.type', value: 'single-node' }
 when:
   - elastic_component == 'elasticsearch'
 notify: Restart Elasticsearch
```

```
- name: Configure Kibana server host
 lineinfile:
   path: /etc/kibana/kibana.yml
   regexp: '^server.host:'
   line: 'server.host: "0.0.0.0"'
 when:
   - elastic_component == 'kibana'
 notify: Restart Kibana

- name: Configure Kibana Elasticsearch host
 lineinfile:
   path: /etc/kibana/kibana.yml
   regexp: '^elasticsearch.hosts:'
   line: 'elasticsearch.hosts: ["http://127.0.0.1:9200"]'
 when:
   - elastic_component == 'kibana'
 notify: Restart Kibana

- name: Configure Logstash output
 copy:
   dest: /etc/logstash/conf.d/02-beats-input.conf
   content: |
    input {
      beats {
        port => 5044
      }
```

```
    }
    output {
      elasticsearch {
        hosts => ["http://127.0.0.1:9200"]
      }
    }
   owner: root
   group: root
   mode: '0644'
 when:
   - elastic_component == 'logstash'
 notify: Restart Logstash


- name: Enable and start requested Elastic service
 systemd:
   name: "{{ elastic_component }}"
   enabled: yes
   state: started
```

Nagios for ubuntu Task:

```
- name: Install Nagios packages on Ubuntu host
 apt:
```

```
    name: "{{ nagios_packages_ubuntu }}"
    state: present
    update_cache: yes
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Nagios on Ubuntu
  systemd:
    name: nagios4
    enabled: yes
    state: started
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Apache for Nagios UI on Ubuntu
  systemd:
    name: apache2
    enabled: yes
    state: started
  when: ansible_facts['distribution'] == 'Ubuntu'
```

2.3:
Lamp Task:

```
- name: Install Apache and PHP packages on Ubuntu web host
  apt:
    name: "{{ lamp_web_packages_ubuntu }}"
    state: present
    update_cache: yes
```

```
 when:
   - lamp_component == 'web'
   - ansible_facts['distribution'] == 'Ubuntu'

- name: Install Apache and PHP packages on CentOS web host
 dnf:
   name: "{{ lamp_web_packages_centos }}"
   state: present
   update_cache: yes
 when:
   - lamp_component == 'web'
   - ansible_distribution == 'CentOS'

- name: Create PHP test page
 copy:
   dest: /var/www/html/index.php
   content: |
    <?php
    phpinfo();
    ?>
   owner: root
   group: root
   mode: '0644'
 when:
   - lamp_component == 'web'
   - ansible_facts['distribution'] in ['Ubuntu', 'CentOS']
```

```yaml
- name: Enable and start Apache web service on Ubuntu
 systemd:
   name: apache2
   enabled: yes
   state: started
 when:
   - lamp_component == 'web'
   - ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Apache web service on CentOS
 systemd:
   name: httpd
   enabled: yes
   state: started
 when:
   - lamp_component == 'web'
   - ansible_distribution == 'CentOS'

- name: Install MariaDB on Ubuntu database host
 apt:
   name: "{{ lamp_db_packages_ubuntu }}"
   state: present
   update_cache: yes
 when:
   - lamp_component == 'db'
```

```
    - ansible_facts['distribution'] == 'Ubuntu'


- name: Enable and start MariaDB on Ubuntu database host
  systemd:
    name: mariadb
    enabled: yes
    state: started
  when:
    - lamp_component == 'db'
    - ansible_facts['distribution'] == 'Ubuntu'


- name: Install MariaDB on CentOS database host
  dnf:
    name: "{{ lamp_db_packages_centos }}"
    state: present
    update_cache: yes
  when:
    - lamp_component == 'db'
    - ansible_distribution == "CentOS"


- name: Enable and start MariaDB on CentOS database host
  systemd:
    name: mariadb
    enabled: yes
    state: started
  when:
```

```
  - lamp_component == 'db'

  - ansible_distribution == "CentOS"
```

Result:
TAKE NOTE: I opened centos first because the pc i'm using can't handle both running… so here i started with centos
Centos:

```
*******************************************************
centos_host                : ok=5    changed=1    unreachable=0    failed=1    skipped=7    rescued=0
  ignored=0
```

The task was almost okay, fixing it…
Error was the startup issue of the centos:

```
isaac@Workstation:~/CPE212_Alegre/CPE212_Midterm_Exam$ ansible-playbook midterm.yml -K
TASK [lamp : Enable and start MariaDB on Ubuntu database host] ************************************
skipping: [centos_host]

TASK [lamp : Install MariaDB on CentOS database host] *********************************************
changed: [centos_host]

TASK [lamp : Enable and start MariaDB on CentOS database host] ************************************
changed: [centos_host]

PLAY RECAP ****************************************************************************************
centos_host                : ok=32   changed=15   unreachable=0    failed=0    skipped=39   rescued=0
  ignored=0
ubuntu_host                : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0
  ignored=0
```

Ubuntu:

```
PLAY RECAP ****************************************************************************
centos_host                : ok=0    changed=0    unreachable=1    failed=0    skipped=0
    rescued=0    ignored=0
ubuntu_host                : ok=41   changed=18   unreachable=0    failed=0    skipped=4
3   rescued=0    ignored=0

isaac@Workstation:~/CPE212_Alegre/CPE212_Midterm_Exam$
```

Verification:

Centos:

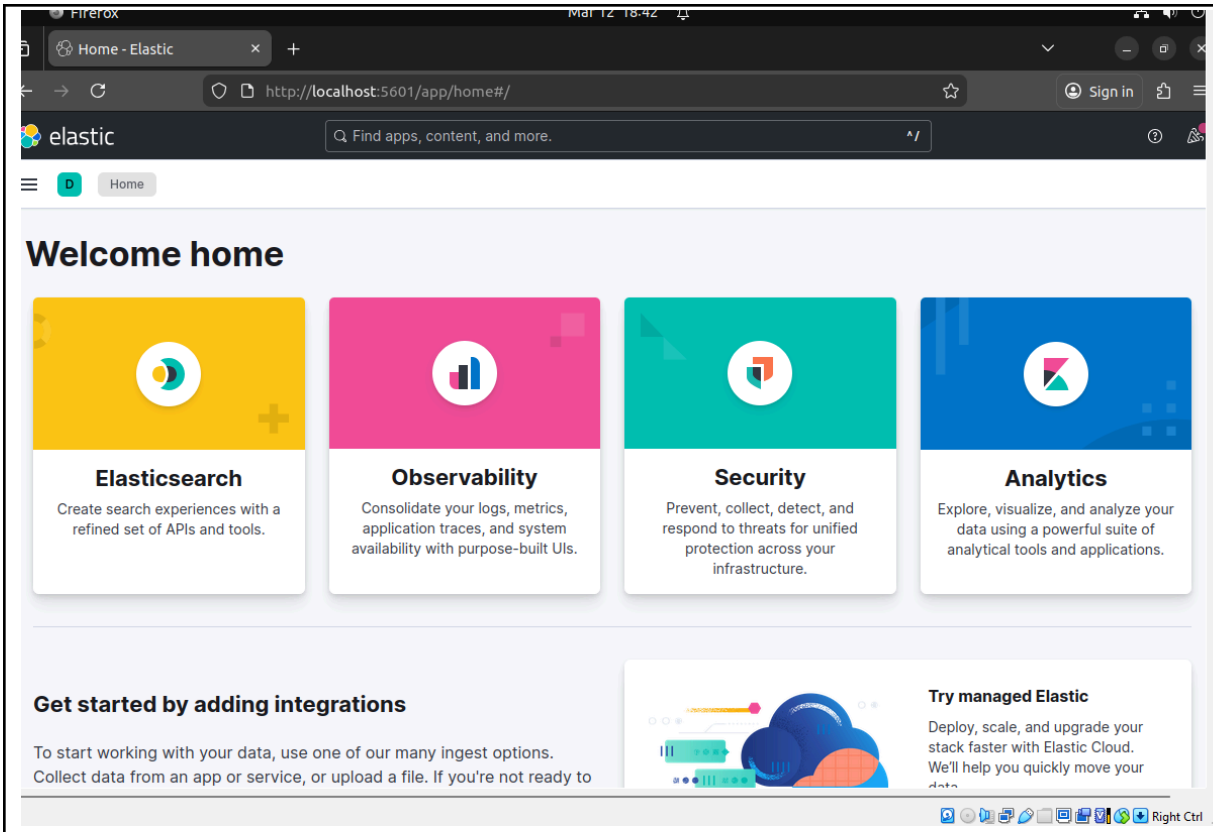
Elastic:

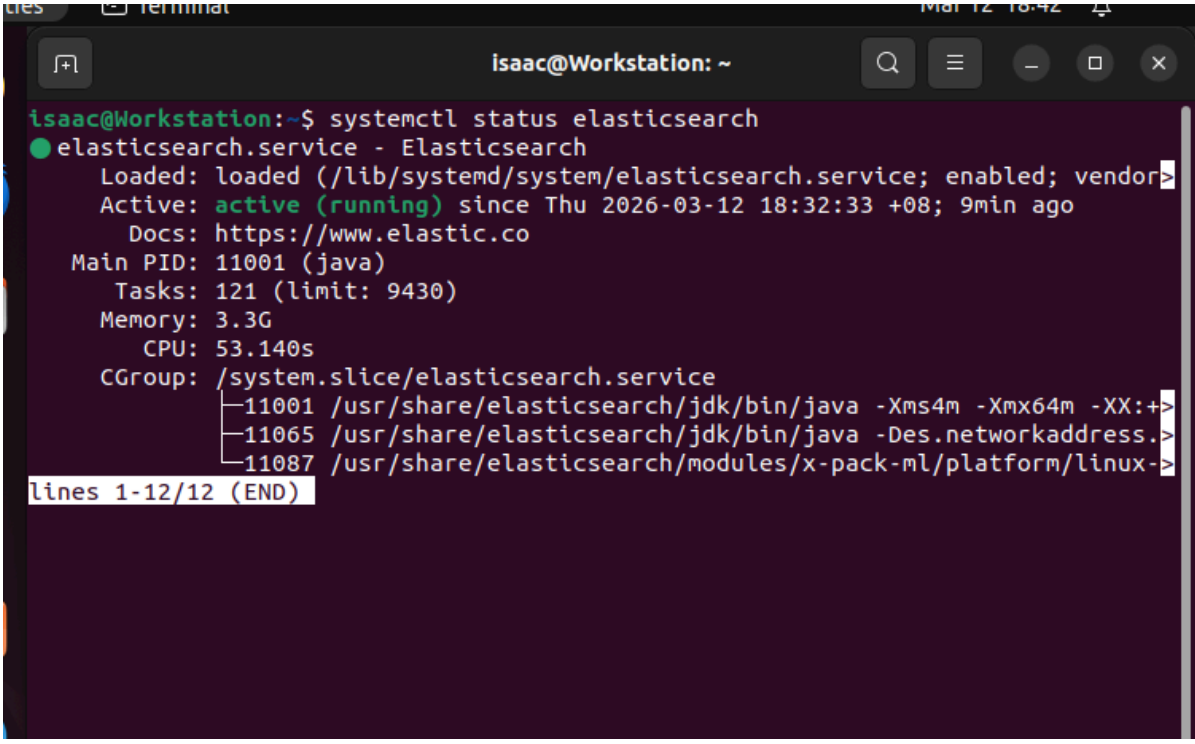
```
[jhonisaacalegre@vbox ~]$ systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
     Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; pr>
     Active: active (running) since Thu 2026-03-12 18:59:49 PST; 43s ago
       Docs: https://www.elastic.co
   Main PID: 1442 (java)
      Tasks: 109 (limit: 58684)
     Memory: 5.1G
        CPU: 47.788s
     CGroup: /system.slice/elasticsearch.service
             ├─1442 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+U>
             ├─2390 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.c>
             └─2416 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x>

Mar 12 18:59:24 vbox systemd[1]: Starting Elasticsearch...
Mar 12 18:59:28 vbox systemd-entrypoint[1442]: WARNING: Unknown module: jdk.int>
Mar 12 18:59:28 vbox systemd-entrypoint[1442]: WARNING: Unknown module: org.ela>
Mar 12 18:59:49 vbox systemd[1]: Started Elasticsearch.
```

Kibana:

```
jhonisaacalegre@vbox:~

[jhonisaacalegre@vbox ~]$ systemctl status kibana
● kibana.service - Kibana
     Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: d>
     Active: active (running) since Thu 2026-03-12 18:59:24 PST; 1min 39s ago
       Docs: https://www.elastic.co
   Main PID: 1444 (node)
      Tasks: 11 (limit: 58684)
     Memory: 757.3M
        CPU: 25.300s
     CGroup: /system.slice/kibana.service
             └─1444 /usr/share/kibana/bin/../node/glibc-217/bin/node /usr/share>

Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.472+08:00][INFO ][plugi>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.472+08:00][INFO ][plugi>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.497+08:00][INFO ][plugi>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.499+08:00][INFO ][plugi>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.516+08:00][INFO ][statu>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.516+08:00][INFO ][statu>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.516+08:00][INFO ][statu>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.516+08:00][INFO ][statu>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.589+08:00][INFO ][plugi>
Mar 12 18:59:56 vbox kibana[1444]: [2026-03-12T18:59:56.596+08:00][INFO ][statu>

[jhonisaacalegre@vbox ~]$
```

Ubuntu:
Elastic:


Home - Elastic
http://localhost:5601/app/home#/
Sign in
elastic
Find apps, content, and more.
Home
Welcome home
Elasticsearch
Create search experiences with a refined set of APIs and tools.
Observability
Consolidate your logs, metrics, application traces, and system availability with purpose-built UIs.
Security
Prevent, collect, detect, and respond to threats for unified protection across your infrastructure.
Analytics
Explore, visualize, and analyze your data using a powerful suite of analytical tools and applications.
Get started by adding integrations
To start working with your data, use one of our many ingest options. Collect data from an app or service, or upload a file. If you're not ready to
Try managed Elastic
Deploy, scale, and upgrade your stack faster with Elastic Cloud. We'll help you quickly move your
Right Ctrl


isaac@Workstation: ~
isaac@Workstation:~$ systemctl status elasticsearch
elasticsearch.service - Elasticsearch
Loaded: loaded (/lib/systemd/system/elasticsearch.service; enabled; vendor
Active: active (running) since Thu 2026-03-12 18:32:33 +08; 9min ago
Docs: https://www.elastic.co
Main PID: 11001 (java)
Tasks: 121 (limit: 9430)
Memory: 3.3G
CPU: 53.140s
CGroup: /system.slice/elasticsearch.service
11001 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+
11065 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.
11087 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-
lines 1-12/12 (END)

Kibana:

```
isaac@Workstation:~$ systemctl status kibana
● kibana.service - Kibana
     Loaded: loaded (/lib/systemd/system/kibana.service; enabled; vendor preset>
     Active: active (running) since Thu 2026-03-12 18:36:37 +08; 11min ago
       Docs: https://www.elastic.co
   Main PID: 12797 (node)
      Tasks: 11 (limit: 9430)
     Memory: 1008.4M
        CPU: 32.636s
     CGroup: /system.slice/kibana.service
             └─12797 /usr/share/kibana/bin/../node/glibc-217/bin/node /usr/shar>
lines 1-10/10 (END)
```

GitHub link:

[CPE212_Alegre/CPE212_Midterm_Exam at main · EngrAlegre/CPE212_Alegre · GitHub]

Conclusions: (link your conclusion from the objective)

In this activity, I have successfully created and designed a robust, automated workflow for managing enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) solution. This process not only standardizes the installation and configuration of availability, performance, and log monitoring agents across all target systems but also ensures that the entire environment is easily reproducible, scalable, and consistently managed, ultimately providing a significant improvement in operational efficiency and reliability over manual methods.